COURSE

EM5120: ECONOMICS FOR TECHNOCRATS

TITLE:

Digital Bookkeeping for Rural Micro-Merchants: An Economic Case Study

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Section 1: Problem Identification and Relevance

1.1 The problem

India's rural economy runs on small enterprises. The government's 2025 survey of unincorporated non-agricultural enterprises (MoSPI) counts 7.92 crore establishments, and rural entrepreneur's use of the internet for business has risen to 31.06%. Payments have digitised fastest: UPI is widely accepted, and since small merchants receiving up to ₹1 lakh a month pay zero merchant discount rate, there is little cost to accepting it (NPCI, 2026).

But payment digitisation has not become **management digitisation**. A rural retailer may receive a UPI payment in seconds and still track what customers owe, what is running low, and what she earned this week from memory or a paper notebook. This creates four costs:

1. **Credit leakage:** unrecorded or forgotten customer credit is never collected **Poor stocking decisions:** stockouts and dead stock without sales records.
2. **Invisibility to lenders:** without records, a lender cannot cheaply tell a good borrower from a bad one, so credit is rationed or priced high.
3. **Time cost:** manual calculation takes hours the owner could spend selling.

The problem persists because the market doesn't fix it on its own. Merchants undervalue record-keeping (it feels like overhead), and past attempts struggled to monetise them directly.

1.2 The proposed solution

A **local-language, voice-assisted bookkeeping and inventory service** for low-volume merchants, designed around low literacy and basic phones:

- Record sales, credit given and expenses by voice or one-tap entry.
- Optionally import UPI receipts so digital sales are recorded automatically.
- Give a short weekly spoken summary: who owes money, what is running low, and this week's estimated profit.
- Light on data and storage, with no recurring fee in the base design.

The design choice is deliberate. Evidence from a randomised experiment in Nairobi found that a free POS app with automated analytics raised daily record-keeping by 7 to 9 percentage points

over the same app without analytics (IGC), while a Colombian pilot found plain record-keeping apps were not widely adopted (IPA). The value lies in turning records into insight, not in recording alone.

1.3 Target customer

- **Primary user:** rural small retailers with 0 to 2 workers (general stores, tea stalls, small suppliers) who already accept UPI.
- **Segmentation:** merchants receiving up to ₹1 lakh a month (exempt from MDR) versus larger ones who may now bear 0.4% MDR on specified payments above ₹2,000 from October 2026 (NPCI, 2026). The second group has more reason to track margins.
- **Secondary customer (the payer):** an institutional partner such as a bank, NBFC, distributor or self-help group federation that benefits from better merchant records.

1.4 Why it is worth solving: the economic value created

Value accrues to three parties:

- **Merchant:** recovered credit, margin gained from better stocking, and time saved valued at the local wage rate, plus easier access to credit once records exist.
- **Lender or distributor:** lower cost of assessing borrowers and lower expected loss, which reduces information asymmetry.
- **Society:** higher productivity and a stronger move toward formal, traceable business activity.

Considering merchant value as :

V = recovered credit + margin gain from stock decisions + time saved × wage + value of credit access.

Policy context strengthens the case. The Income-tax Act, 2025 keeps a 6% presumptive rate for digital receipts against 8% for cash, and it waives books for presumptive taxpayers but requires records if income is declared below those rates. That means the regulatory push to keep books is weak for tiny merchants, so any adoption has to come from the product's own value or from a partner's interest.

Term	Assumption	Value per month

Recovered credit	₹20,000 credit sales, 5% unrecovered, app recovers one-third	≈ ₹330
Margin from better stocking	Fewer stockouts and dead stock	≈ ₹300
Time saved	4 hours × ₹50 an hour	₹200
Total V		**≈ ₹830**

1.5 The key decision and alternatives

Decision: Should I launch this service, and if so, should I sell it to merchants directly or fund it through an institutional partner?

Option	What it means	Who pays	Core trade-off
A. Do not launch	Maintain the status quo; invest nowhere	None	Zero risk and zero social value; forgoes the opportunity
B. Direct subscription	Merchants pay a monthly fee	Merchant	Simple revenue, but low willingness to pay and free competitors
C. Freemium	Free basic tier, paid premium features	A small share of merchants	Large user base needed; conversion rate drives viability
D. Partner-funded	Free to merchants; a bank, lender or distributor pays	Institution	Higher revenue per merchant, but dependence on partner and channel

E. Integrate into an existing payments app	Offer as a feature, not a standalone product	Platform	Lower cost and instant reach, but little control and thin margins

The remaining sections test these options against demand, costs and uncertainty to choose one.

Section 2: Demand, Market and Competitive Environment

Market definition. The relevant market is rural micro-retailers and small traders with 0 to 2 workers who already accept UPI. MoSPI's 2025 survey shows 31.1% of rural establishments use the internet for business but only 3.0% use a computer, so the product must be phone-first. Own-account enterprises with no hired workers are far less connected than those with workers (34.2% versus 72.7% nationally). The addressable base is therefore a fraction of India's 7.34 crore unincorporated establishments.

Customers, competitors and substitutes. Customers are merchants and, as payers, institutions such as banks, NBFCs and distributors. Direct competitors are Khatabook, OkCredit, while payments apps like PhonePe and Paytm bundle bookkeeping free. The strongest substitute is the paper notebook.

Demand and willingness to pay. Price sensitivity is high. OkCredit lists plans at ₹30, ₹75 and ₹99 a month, so ₹30 is an observed price point, yet monetisation has been weak. Khatabook was pre-revenue until FY20, and OkCredit earned about ₹3.79 lakh against ₹114.6 crore spent in FY21 (press-reported, older). OkCredit now claims 2 lakh paying shopkeepers (self-reported, unverified). I therefore assume only a small minority pay **[A7]**.

Differentiation, structure and barriers. My edge is voice-first, vernacular insight, since analytics raise record-keeping more than plain logging. The market resembles monopolistic competition: differentiated products, free tiers and low technical barriers. The real barriers are customer acquisition, trust and switching costs once merchants have entered data. From 15 October 2026, merchants above ₹1 lakh a month may face 0.4% MDR on specified payments above ₹2,000 (NPCI), giving them more reason to track margins.

Likely responses. Incumbents could copy voice features quickly, and a payments app could bundle free analytics, collapsing paid demand. That favours partner funding, where merchants pay nothing, though partners may then push fees down or build in-house tools.

Section 3: Economics of the Proposed Solution

3.1 Cost structure

- **Fixed (monthly):** team ₹6 lakh, platform and voice-model base ₹1 lakh, partnerships and onboarding ₹1 lakh. Total ₹8 lakh **[A1]**. These costs are step-fixed, so I assume they rise as scale grows **[A2]**.
- **Variable and marginal (per active merchant per month):** cloud ₹3, speech processing ₹6, notifications ₹2, support ₹4. Total ₹15 **[A3]**. This is also the marginal cost of one more merchant while capacity lasts. Free-tier users cost less, ₹5, because the voice features are limited **[A4]**.
- **Sunk once spent:** ₹40 lakh to build the product **[A5]**.
- **Opportunity costs:** the founder's alternative earnings are included in team cost at market salary, and the cost of capital is captured by a 15% discount rate **[A6]**.

3.2 Revenue models and evidence

Payment commissions are excluded as a revenue line: small merchants pay zero MDR under the new framework (NPCI, 2026), so there is no fee to share. Revenue comes from one of three models, tested below.

3.3 Analysis 1: Contribution and break-even per model

Break-even merchants = monthly fixed cost ÷ contribution per merchant, using ₹8 lakh fixed.

Model	Revenue assumption	Contribution per merchant	Break-even
B. Direct subscription	₹30/month, all users pay **[A7]**	₹30 − ₹15 = ₹15	53,333 paying merchants
C. Freemium	8% pay ₹60, others free **[A7]**	50p − 5 = **−₹1** at p = 8%	Needs conversion p ≥ **10%**. At 15%: 3,20,000 users. At 20%: 1,60,000 users
D. Partner-funded	Partner pays ₹40 per active merchant per month **[A8]**	₹40 − ₹15 = ₹25	32,000 merchants (80,000 at ₹25 fee, 20,000 at ₹55 fee)

Freemium contribution per average user is $p \times (60 - 15) - (1 - p) \times 5 = 50p - 5$.

Reading the result: direct models need many users to cover fixed costs and depend on a small share converting, which fits the low willingness to pay seen at Khatabook and OkCredit. Partner funding reaches break-even at a lower scale because the payer values the merchant's data more than the merchant does.

3.4 Analysis 2: Economies of scale

Average cost per merchant per month = ₹15 + ₹8 lakh ÷ merchants:

- 10,000 merchants: ₹95
- 50,000 merchants: ₹31
- 1,00,000 merchants: ₹23

Costs fall steeply with scale, supporting a shared-platform, partner-distribution model.

Assumption register

ID	Assumption	Justification
A1	Fixed cost ₹8 lakh a month	Small team of about 6 at ₹1 lakh loaded, plus platform and onboarding
A2	Fixed costs rise to ₹10, 12 and 12.5 lakh a month in years 2 to 5	Support and partnership teams scale
A3	Variable cost ₹15 per merchant	Component estimates of cloud, speech, messaging and support
A4	Free-tier user costs ₹5	Voice features limited to paid tier
A5	Build cost ₹40 lakh	MVP in several languages
A6	15% discount rate	Early-stage venture risk
A7	₹30 subscription; 8% freemium conversion at ₹60	Low willingness to pay seen in Khatabook and OkCredit history
A8	Partner fee ₹40 (₹25 to ₹55 range)	Derived from partner value in A9
A9	1.5% loss reduction on ₹40,000; ₹240 underwriting saving	Illustrative lender economics
A10	Merchant ramp of 10k to 1.2 lakh	Depends on a partner's distribution

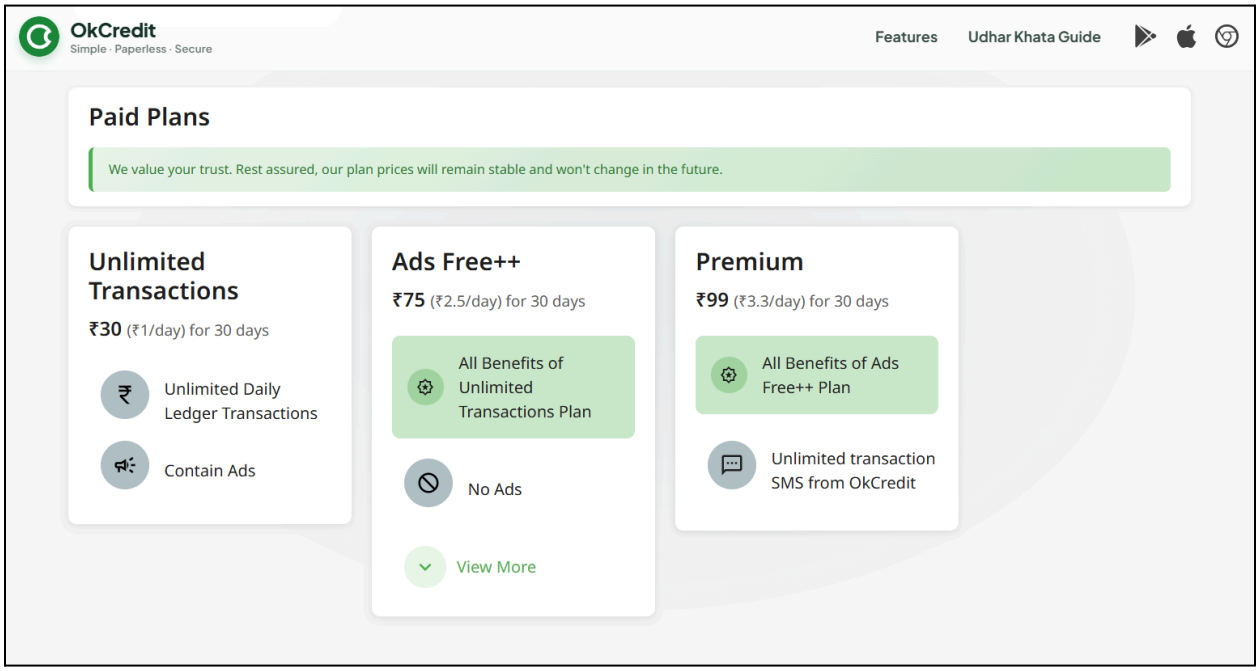
3.5 Analysis 3: Five-year investment appraisal (Option D, base case)

Assumptions: average active merchants (net of churn) of 10,000 / 40,000 / 90,000 / 1,10,000 / 1,20,000 in years 1 to 5 **[A10]**; fixed costs of ₹96 / 120 / 144 / 150 / 150 lakh a year **[A2]**; no tax, inflation or working capital; 15% discount rate.

Year	Contribution (₹ lakh)	Fixed (₹ lakh)	Cash flow (₹ lakh)
0			−40
1	30	96	−66
2	120	120	0
3	270	144	+126
4	330	150	+180
5	360	150	+210

- **Base NPV ≈ +₹1.93 crore.** Undiscounted payback comes in year 3 (cumulative −₹106 lakh at the end of year 2).
- **Peak funding need ≈ ₹1.06 crore.**
- **Low case** (fee ₹25, half the merchants, same fixed costs): NPV ≈ **−₹3.37 crore**.
- **High case** (fee ₹55, 1.5× merchants): NPV ≈ **+₹11.2 crore**.

3.6 Evidence



***Source**:* OkCredit website, paid plans page.
Three listed plans for 30 days: Unlimited Transactions at ₹30 (with ads), Ads Free++ at ₹75, and Premium at ₹99 (adds unlimited transaction SMS from OkCredit).
Use: Direct market evidence for the subscription price in **A7**. A ₹30 plan from the best-known incumbent supports using ₹30 as the base case, and ₹75 to ₹99 show what premium features sell for.



Plan	Starter	Creator	RECOMMENDED Professional	Scale	Enterprise
Monthly subscription	₹1,000	₹2,500	₹7,500	₹35,000	₹1,00,000
Annual subscription	₹10,000	₹25,000	₹75,000	₹3,50,000	₹10,00,000
Annual saving	₹2,000	₹5,000	₹15,000	₹70,000	₹2,00,000
Credits per Monthly term	1,000	2,500	7,500	35,000	1,00,000
Credits per Annual term (upfront)	12,000	30,000	90,000	4,20,000	12,00,000
Commercial use	Included	Included	Included	Included	Included
API key access	Included	Included	Included	Included	Included
Speech recognition	0.55/min	0.50/min	0.45/min	0.38/min	0.33/min
Voice generation	0.95/1K characters	0.85/1K characters	0.75/1K characters	0.60/1K characters	0.50/1K characters
Translation	0.55/1K characters	0.50/1K characters	0.45/1K characters	0.38/1K characters	0.33/1K characters
Video dubbing / target language	33.33/min	27.78/min	25.00/min	23.33/min	22.22/min
Transliteration	0.55/1K characters	0.50/1K characters	0.45/1K characters	0.38/1K characters	0.33/1K characters
Cloud OCR	1/page or image	1/page or image	1/page or image	1/page or image	1/page or image
Support	Docs & samples	Docs & samples	Email	Call + engineer	AI expert + SLA

Source: Bhashini.ai pricing page.

What it shows: Plans from ₹1,000 to ₹1,00,000 a month. Speech recognition costs ₹0.55 per minute (Starter) down to ₹0.33 (Enterprise). Voice generation costs ₹0.95 down to ₹0.50 per 1,000 characters. Commercial use and API access are included.

Speech cost (A3): assuming 10 minutes of speech recognition a month at ₹0.45 per minute (Professional plan) and four spoken weekly summaries of about 300 characters at ₹0.75 per 1,000 characters gives ≈ ₹5.4, rounded to ₹6. This reads credits as rupees (1 credit = ₹1), which I assume from the page. Supplier price falls to ₹0.33 per minute at Enterprise volume, which supports the scale economies in Section 3.4.

Statement 16: Percentage of establishments 'using computer' and 'using internet' during last 365 days

BAC	Establishment type	Percentage of establishments using computer			Percentage of establishments using internet		
		OAE	HWE	All	OAE	HWE	All
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Rural	Manufacturing	0.1	6.8	0.5	17.3	57.3	19.6
	Trade	1.1	22.1	2.6	40.1	74.3	42.7
	Other services	3.8	19.6	5.1	29.3	57.9	31.6
	All*	**1.9**	**17.3**	**3.0**	**28.6**	**62.6**	**31.1**
Urban	Manufacturing	0.7	12.9	3.1	27.2	69.5	35.3
	Trade	2.7	25.1	8.4	54.2	78.2	60.3
	Other services	6.2	36.0	11.4	40.8	79.4	47.5
	All*	**3.9**	**26.6**	**8.5**	**41.8**	**76.8**	**48.9**
Rural +Urban	Manufacturing	0.3	11.0	1.5	20.9	65.7	25.8
	Trade	1.8	24.4	5.5	46.5	77.3	51.6
	Other services	4.9	30.5	8.2	34.5	72.2	39.2
	All*	**2.7**	**23.9**	**5.6**	**34.2**	**72.7**	**39.4**

**All includes Non-captive electricity generation* *Reference: table 20 of Appendix A*

***Source**:* MoSPI survey report, section 3.6 and Statement 16, covering January to December 2025 per the text.

39.4% of all establishments used the internet for business, versus 72.7% of establishments with hired workers and 34.2% of own-account enterprises. Rural figures are 31.1% overall and 42.7% in trade (own-account trade 40.1%). Computer use is very low in rural areas: 3.0% overall and 2.6% in trade.

Use: Market size and adoption for Section 2, and support for a **phone-first, no-computer** design.

Section 4: Strategic Choice and Uncertainty

Comparison (base case from Section 3). Option A (do not launch) has NPV of zero and no risk, but gives up the opportunity. Option B (subscription) needs 53,333 paying merchants, which is doubtful given low willingness to pay. Option C (freemium) works only if conversion

reaches 10% or more. Option D (partner-funded) breaks even at 32,000 merchants, with base NPV of about +₹1.93 crore. Option E (feature inside a payments app) needs little capital but gives up control and margin (qualitative only, not modelled).

Sensitivity (Option D).

- Half the merchants and a ₹25 fee: NPV falls to **−₹3.37 crore**.
- Variable cost rises by ₹5: contribution drops to ₹20 and break-even rises to 40,000 (+25%).
- D survives only if the partner fee stays above about ₹33; a price war on bookkeeping would pressure that fee.
- Cheaper speech technology lowers my costs but also lowers barriers for entrants.

Game-theoretic reasoning. Treat my choice (charge merchants vs partner-funded) against the incumbent's (bundle free vs not). If the incumbent bundles, charging merchants earns a negative payoff, and if it doesn't, a modest one. Partner funding earns positive payoffs either way. So direct charging is weakly dominated and D is the robust choice. The main strategic risk is hold-up: once I have sunk the ₹40 lakh build cost, a single partner can renegotiate the fee. Mitigations are multiple partners and minimum-volume commitments.

My strategy \ Incumbent	Bundles free	Does not bundle
Charge merchants (B or C)	negative (≈ −0.4)	small positive (≈ +0.3)
Partner-funded (D)	positive if fee holds above ₹33	+1.93

Implication. Because demand and partner willingness to pay are uncertain, a staged pilot has option value: commit a fraction of the build cost, then scale only if partner and retention targets are met.

Section 5: Decision and Reflection

Decision: Modify. I do not recommend launching a standalone, merchant-paid product (Options B or C). I recommend a **stage-gated, partner-funded pilot (Option D)**, with integration into an existing payments app (Option E) as the fallback if no partner commits.

Why. The Section 3 analysis shows that charging merchants needs about 53,333 paying users (B) or at least 10% conversion (C), which is hard to justify given the low willingness to pay seen at Khatabook and OkCredit. Zero MDR for small merchants removes payment commissions as a revenue line (NPCI, 2026). Partner funding breaks even at about 32,000 merchants at a ₹40 fee, with a base NPV of about +₹1.93 crore. It is also the strategy that survives a free bundled competitor (Section 4). The product design rests on evidence that analytics raise record-keeping more than plain logging (IGC, Nairobi).

Staging. Spend about ₹10 lakh of the ₹40 lakh build cost on a pilot , and scale only if: (1) a partner signs a commitment covering about 32,000 merchants at a fee of ₹40 or more, and (2) pilot retention meets the churn assumed in A10. This limits sunk-cost exposure and gives the partner less room to renegotiate after I have invested.

What would change my decision.

- A partner fee below about ₹33 turns the base-case NPV negative, so I would negotiate a floor price of at least ₹40 and treat any fee below ₹33 as grounds to reject or move to Option E.
- Evidence that merchants will pay ₹30 or more with conversion above 10% would favour **direct freemium**.
- A free, vernacular analytics feature from a major payments app would favour **Option E**.
- Pilot retention well above assumptions would favour **scaling** faster.

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